

PRANEETH REDDY ANKEY

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Professional Summary

Full Stack & GenAI Engineer with 1+ year of production experience building agentic RAG systems, AI-powered pipelines, and real-time web applications using Python, React.js, TypeScript, and FastAPI. Skilled in LLM integration (Llama 4, Groq SDK, OpenAI-compatible APIs), vector databases (Supabase pgvector), prompt engineering, and automated document-processing workflows. Proven delivery across defence (DRDO) and enterprise environments. Passionate about fusing generative AI with scalable full-stack development to ship user-centric products.

Technical Skills

Languages: Python, JavaScript (ES6+), TypeScript, SQL, Java, C, C++

AI / GenAI: Llama 4, Groq SDK, OpenAI API, Pydantic AI, LangChain, LlamaIndex, Crawl4AI, Nanonets OCR, Prompt Engineering, Text Embeddings, Chunking Strategies, RAG Pipelines, Agentic AI, AI Summarization Workflows

Frontend: React 19, React.js, Next.js, Tailwind CSS, Vega-Lite, HTML5, CSS3

Backend: FastAPI, Node.js, RESTful APIs, WebSockets, JWT / OAuth 2.0

Databases / Vector Stores: Supabase pgvector, PostgreSQL, SQL, Pinecone (familiar)

Data Processing: Arquero, PyMuPDF, openpyxl, Hugging Face Transformers, Web Scraping

Tools & DevOps: Docker, GitHub Actions (CI/CD), Git, AWS S3 / GCP Cloud Run, Microsoft Teams API, VS Code

Concepts: Semantic Search, Hierarchical Data Modelling, Multi-Agent Systems, Streaming LLM Responses, Vector Search

Work Experience

Defence Research & Development Organisation (DRDO)

Hyderabad, India

Project Intern — Full Stack Developer

Nov 2023 – May 2024 (213 Days)

- Led end-to-end development of a **Live Missile Data Simulation** system, owning both front-end and back-end modules across the full 213-day project lifecycle.
- Built real-time missile trajectory dashboards in **React.js** with dynamic, responsive UI components rendering live simulation data streams.
- Developed Python/**FastAPI** backend services for high-throughput data ingestion, processing, and automated mission analysis report generation, ensuring reliable server-client communication.
- Designed a **4-level hierarchical data model** (Main Category → Sub Category → Time-based grouping → Metrics) for the NOC Dashboard, enabling faster graph rendering and efficient filtering of large-scale Excel-sourced datasets.

Cognitbotz

Hyderabad, India

Software Development Intern

170 Days

- Designed and delivered **Meet-Ops**, an autonomous meeting bot that joins Microsoft Teams calls, captures transcripts, and surfaces AI-driven insights via a **React.js** web dashboard.
- Built an end-to-end **AI summarization pipeline** transforming raw transcripts into actionable meeting insights, integrated with backend REST APIs.
- Deployed cloud-integrated services using **Docker, PostgreSQL, and Microsoft Teams API**; implemented **JWT-based authentication** for secure persistent data storage.
- Set up a **CI/CD pipeline** using GitHub Actions to automate containerised deployments, reducing manual deployment overhead.

Projects

OpenViz — Generative AI Analytics Platform | React 19, TypeScript, Vega-Lite, Llama 4, Groq SDK, Arquero

- Architected a React 19 & Vega-Lite visualization engine with a hybrid drag-and-drop / natural-language interface powered by **Llama 4 via Groq SDK**, enabling prompt-driven chart generation.
- Implemented a **client-side RAG pipeline** with Arquero for instant data profiling and context-aware chart suggestions, with **text embeddings and chunking strategies** ensuring zero-latency, privacy-preserving interactions.
- Applied **prompt engineering** techniques to optimise LLM output structure for reliable Vega-Lite spec generation.

AI-Powered Agent Builder (Agentic RAG) | Pydantic AI, LangChain, Crawl4AI, Supabase pgvector, Python

- Architected a conversational AI system generating production-ready Pydantic AI agents, achieving an **85% success rate** for synthesised agent code.

- Built an automated ingestion pipeline using **Crawl4AI** to scrape and index documentation into **Supabase pgvector** with optimised chunking strategies, enabling high-fidelity semantic search for context retrieval.
- Integrated **LangChain** orchestration layer for multi-step reasoning and tool-use chaining across generated agents.
- Implemented **streaming LLM responses** via the OpenAI-compatible Groq API for low-latency agent interaction.

SAP Invoice Consolidation | *Python, Nanonets OCR, PyMuPDF, Groq API, Prompt Engineering, openpyxl*

- Engineered an AI document-processing pipeline combining **Nanonets OCR** for high-accuracy PDF extraction and **Groq API** with structured prompt engineering for intelligent field parsing, producing SAP-ready Excel output.
- Developed a Tkinter desktop application with multi-page PDF handling, vendor validation, and dynamic tax-code derivation logic to ensure full SAP compliance.

Meet-Ops — Meeting Automation System | *Microsoft Teams API, React.js, TypeScript, Docker, PostgreSQL*

- Built an autonomous meeting assistant bot that joins Teams calls, captures and stores transcripts, and surfaces AI-generated summaries through a connected React dashboard.
- Integrated backend REST APIs with cloud services (Docker Hub, AWS); implemented AI summarisation with **Hugging Face Transformers** for actionable insights.

Live Missile Trajectory Visualisation & Report Generation | *React.js, Python, FastAPI, WebSockets*

- Developed a real-time missile trajectory visualisation system for DRDO, processing live simulation data and rendering it dynamically on a React.js front end via **FastAPI + WebSocket** services.
- Built Python backend pipelines for automated data ingestion, processing, and mission analysis report generation.

Education

Vignan Institute of Technology and Science	Hyderabad, India
<i>Bachelor of Technology — Computer Science and Engineering (CSE)</i>	<i>Graduated 2024</i>
• CGPA: 7.5 / 10	

Telangana State Board	Telangana, India
<i>12th Grade (Intermediate)</i>	<i>2020</i>
• Score: 84.5%	

Certifications

PCAP — Programming Essentials in Python (Cisco / Python Institute)
PSAP — Programming Essentials in Python – Advanced (Python Institute)
CPA — Programming Essentials in C++ (Cisco / C Institute)
CLA — Programming Essentials in C (Cisco / C Institute)